FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects (CVPR 2024 Highlight)

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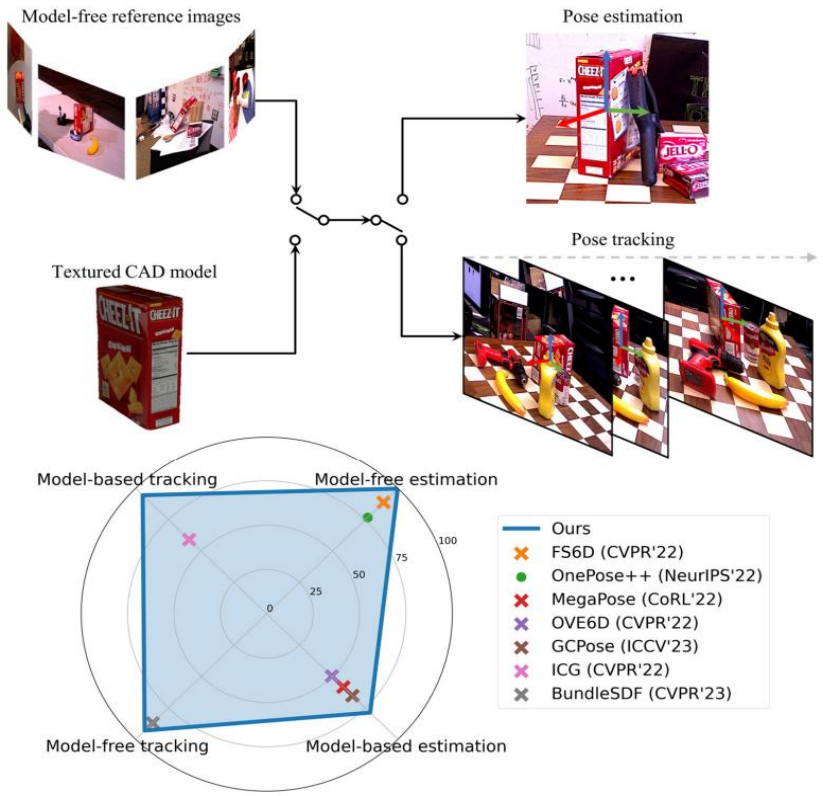
Contents

- Overview
- Introduction
- Preliminaries
- Approach
- Experiment
- Conclusion

Overview

FoundationPose:

a **unified** foundation model for **6D object pose estimation** and **tracking**, supporting both **model-based** and **model-free** setups.



BOP: Benchmark for 6D Object Pose Estimation

HOME CHALLENGES DATASETS LEADERBOARDS SUBMIT RESULTS Sign in

Tasks on seen objects: 6D localization of seen objects 2D detection of seen objects 2D segmentation of seen objects
Tasks on unseen objects: 6D localization of unseen objects 2D detection of unseen objects 2D segmentation of unseen objects

Datasets: Core datasets LM LM-O T-LESS ITODD HB HOPE YCB-V RU-APC IC-BIN IC-MI TUD-L TYO-L

6D localization of unseen objects – Core datasets

This leaderbord shows the overall ranking for Task 4 on the core datasets (LM-O, T-LESS, TUD-L, IC-BIN, ITODD, HB, YCB-V). For each method, the date of the latest considered submission is reported. If more submissions of a method are available for a dataset, the submission with the highest AR_{Core} score is considered. The reported time is the average image processing time averaged over the core datasets.

Show 50 entries Search:

	Date (UTC)	Method	Test image	AR_{Core}	AR_{LM-O}	AR_{T-LESS}	AR_{TUD-L}	AR_{IC-BIN}	AR_{ITODD}	AR_{HB}	AR_{YCB-V}	Time (s)
1	2023-11-19	FoundationPose	RGB-D	0.726	0.733	0.617	0.906	0.528	0.609	0.809	0.882	
2	2023-11-17	PoMZ	RGB-D	0.692	0.684	0.521	0.945	0.503	0.559	0.791	0.840	
3	2023-11-18	SAM6D	RGB-D	0.683	0.687	0.498	0.874	0.561	0.577	0.754	0.828	1.950
4	2023-09-28	GenFlow-MultiHypo16	RGB-D	0.674	0.635	0.521	0.862	0.534	0.554	0.779	0.833	34.578
5	2023-09-25	GenFlow-MultiHypo	RGB-D	0.662	0.622	0.509	0.849	0.524	0.544	0.770	0.818	21.457
6	2023-09-27	Megapose-CNOS_fastSAM+Multihyp_Te...	RGB-D	0.628	0.626	0.487	0.851	0.467	0.468	0.730	0.764	141.965
7	2023-09-27	Megapose-CNOS_fastSAM+Multihyp_Te...	RGB-D	0.623	0.620	0.485	0.846	0.462	0.460	0.725	0.764	116.564
8	2023-09-28	SAM6D-BOP2023-CNOSmask	RGB-D	0.616	0.648	0.483	0.794	0.504	0.351	0.727	0.804	3.872
9	2023-10-18	GenFlow-MultiHypo16-RGB	RGB	0.576	0.572	0.528	0.688	0.458	0.398	0.746	0.642	40.528

Introduction

Background

- **Object pose estimation** is **crucial** for a variety of applications, such as **robotic manipulation** and **mixed reality**.

Instance-level

- **only** work on the **specific object instance** determined at **training time**.
- usually require a **CAD model** for **training** data.

Category-level

- are **limited** to objects **within** the predefined **category** on which they are **trained**.
- **obtaining** category-level **training data** is notoriously **difficult**.

Introduction

To address these limitations,

- more recent efforts have focused on the problem of **instant pose estimation** of **arbitrary novel objects**.
- Progress has been made individually
  - Model-based, where a 3D CAD model is provided
  - Model-free, where reference images is provided.

➔ **Need for a single method to address both setups in a unified way**

Preliminaries

Axes of OPE Research: (무엇을 다루는가)

1. **일반화 수준**
   - Instance-level
   - Category-level
   - Unseen/novel object
2. **3D 모델 의존도**
   - Model-based
   - Model-free
     - Reference View / Reconstruction
3. **RGB / RGB-D**
4. **단일 프레임 / 시퀀스**
   - **Estimation**
   - **Tracking**

+ 물체 속성 (rigid / deformable)
: deformable object는 포즈 자체를 정의하기가
어려워서 다른 문제로 정의되는 편

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➔ 논문들은 주로 이 축들의 조합

like,

6DOPE-GS : novel object + model-free + RGB-D + Tracking

SAM-6D : novel object + model-based + RGB-D + Estimation

FoundationPose : novel object + model-based/free + RGB-D
+ Estimation/Tracking.

Preliminaries

Axes of OPE Research: (어떻게 구하는가)

1. Direct Regression
2. Correspondence-based
3. Template Matching
4. Render-and-Compare
5. Reconstruction-based Optimization
6. Voting / Hough-space

…

연구 Task에 맞는 방법 사용. 조합하여 사용하기도..

Ex)

instance-level + model-based :
template matching + render-and-compare

novel object + model-free :
Reconstruction-based + correspondence

다음에 기회가 되면 각 대표 모델에 대해서 다뤄보도록 하겠습니다.

Method

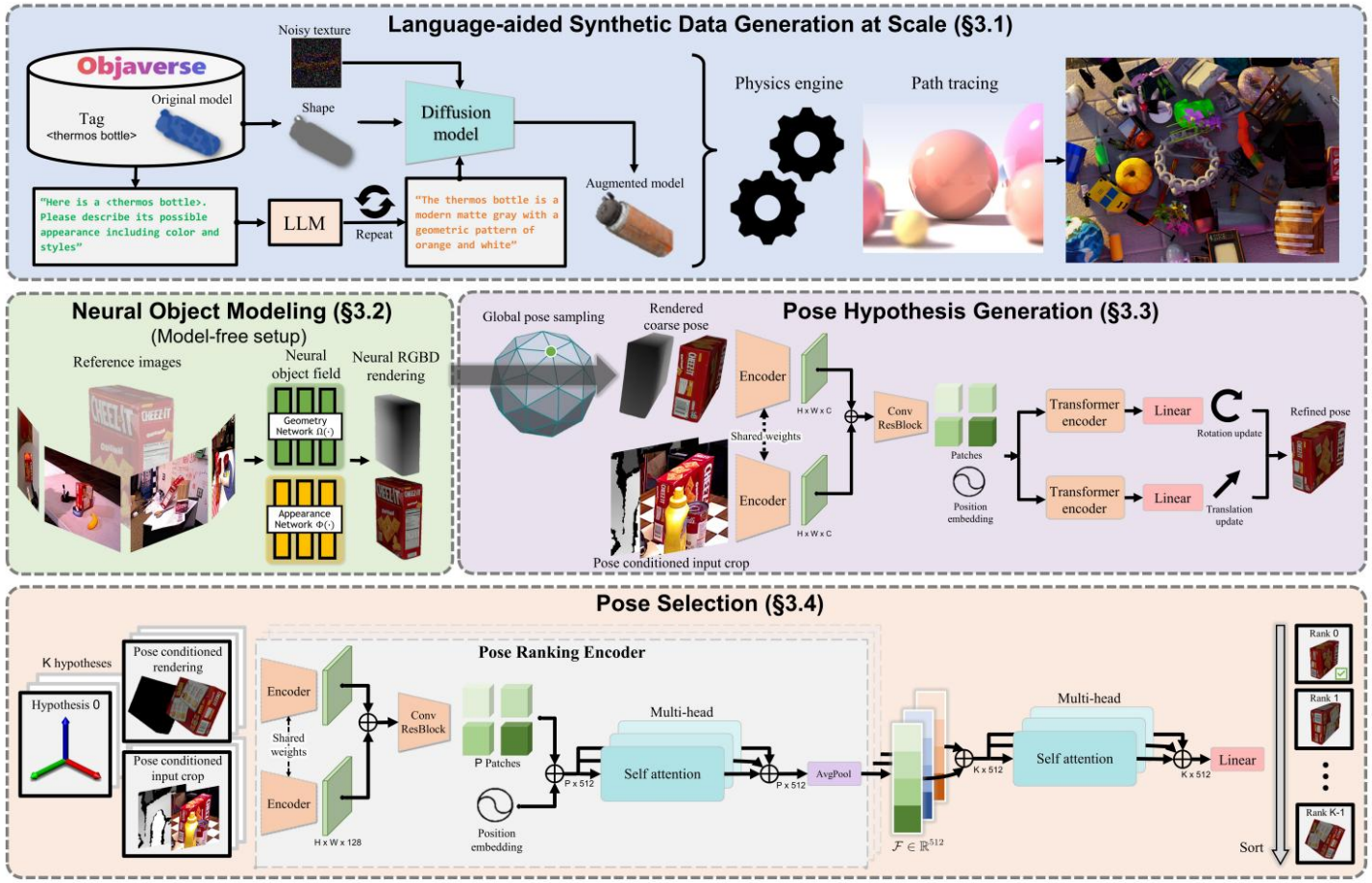
Architecture Overview

3.1. Language-aided Data Generation at Scale

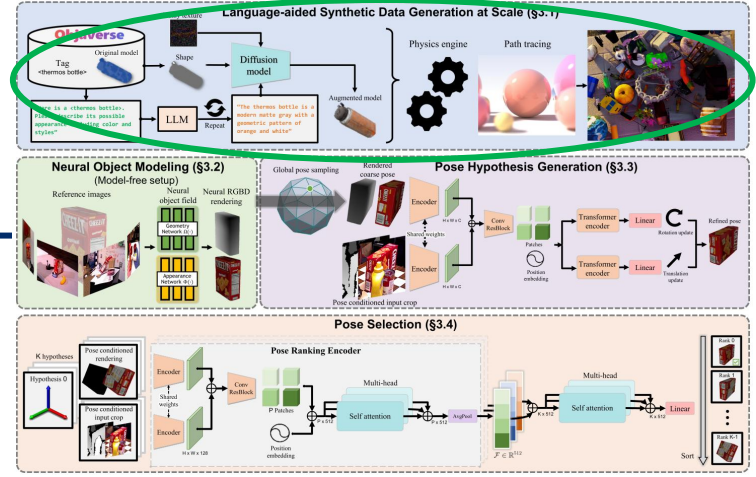
3.2. Neural Object Modeling

3.3. Pose Hypothesis Generation

3.4. Pose Selection



Method



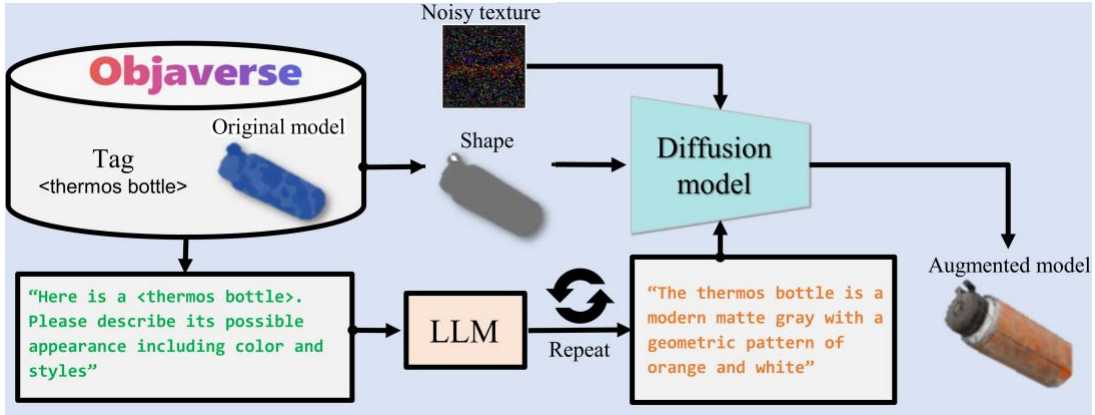
3.1. Language-aided Data Generation at Scale

For strong **generalization**, a **large diversity** of objects and scenes is needed **for training**.

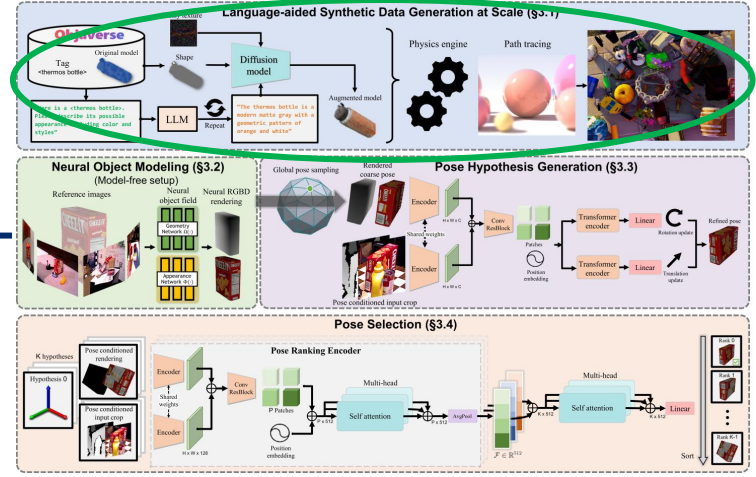
- In the **real world**, obtaining such data is **time-** and **cost-prohibitive**.
- **Synthetic** data **lacks** the **size** and **diversity** in **3D assets**.

Data generation pipeline using

- **3D Assets**
- **LLM**
- **Diffusion**



Method

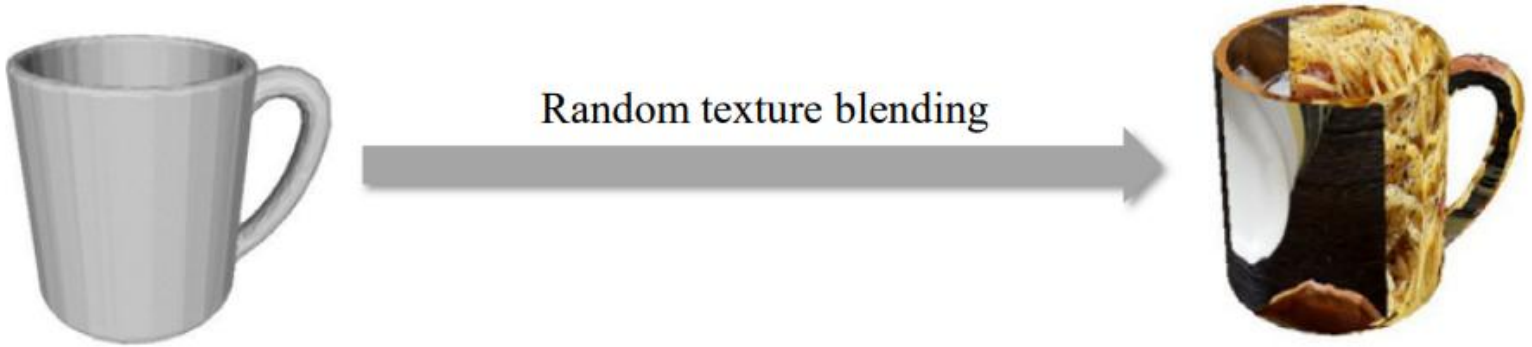


3.1. Language-aided Data Generation at Scale

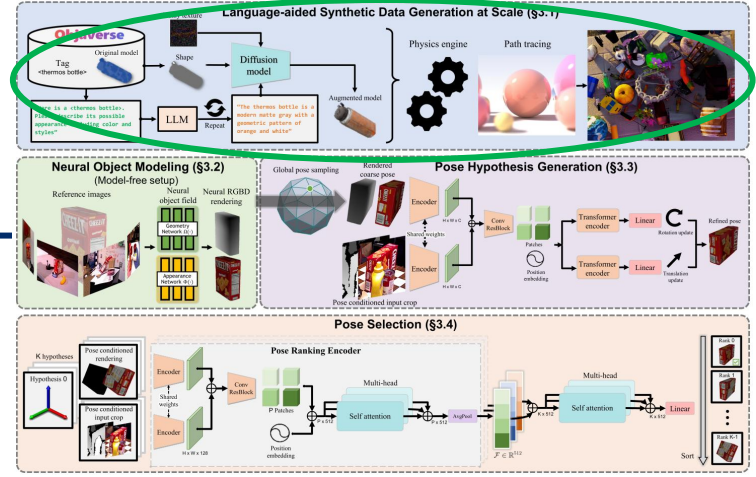
Data generation pipeline in [FS6D]

augment object texture by **randomly pasting images**

→ **unrealistic** results due to the **random mapping**



Method

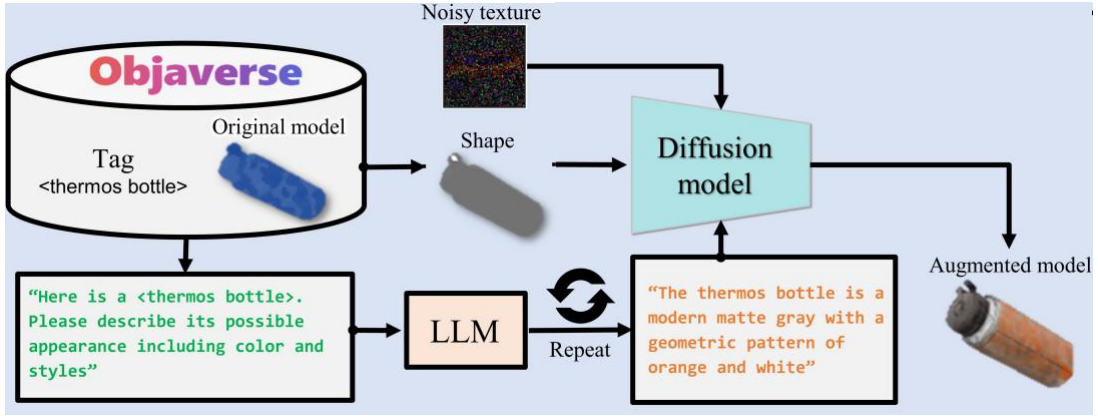


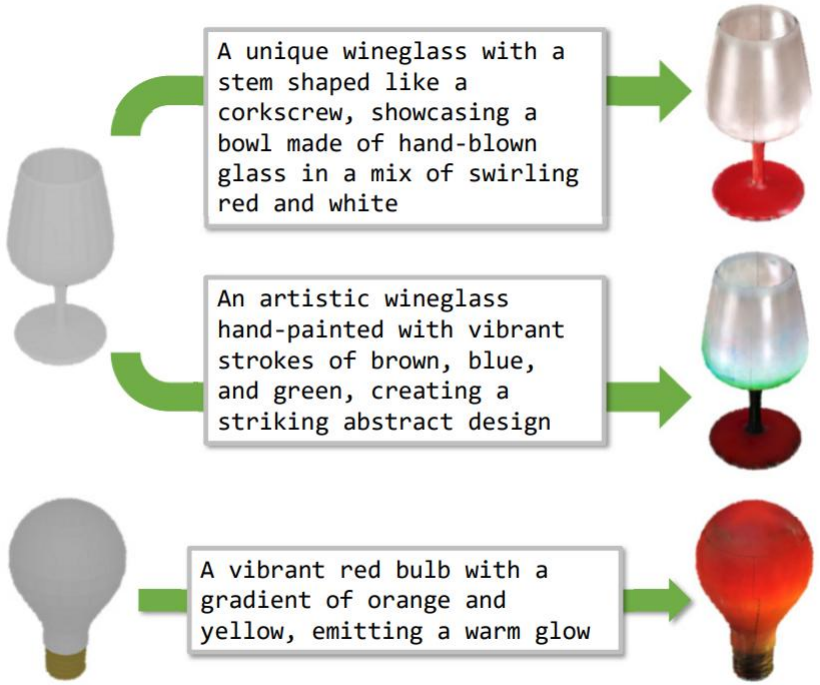
3.1. Language-aided Data Generation at Scale

Data generation pipeline in [FoundationPose]

Objaverse, ChatGPT, TexFusion

- Enables full **automation**.





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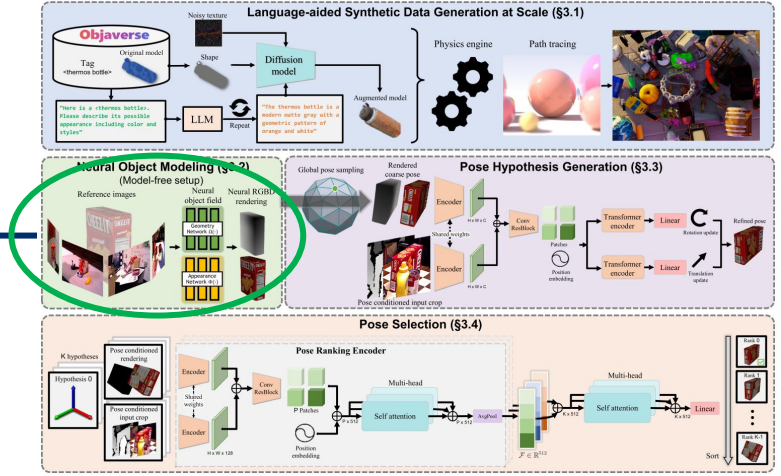
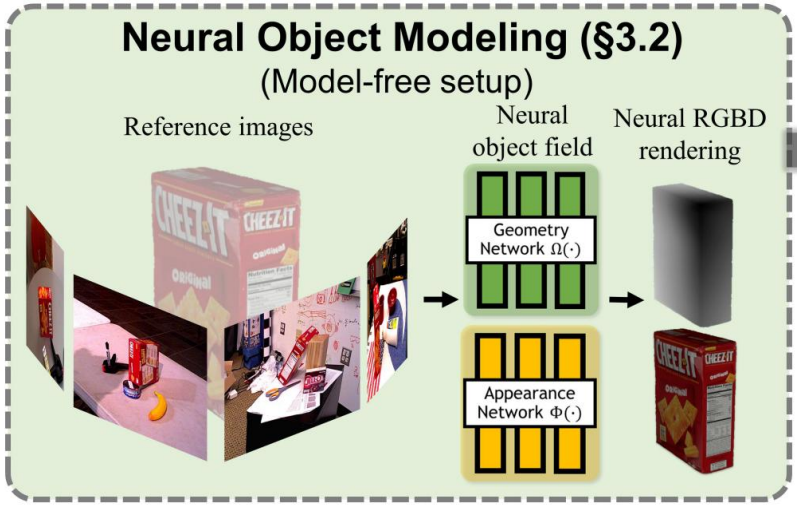


3.2. Neural Object Modeling (For model-free)

Field Representation.

- **Geometry function** $\Omega : x \mapsto s$
  - Input: 3D point $x \in \mathbb{R}^3$
  - Output: a signed distance value $s \in \mathbb{R}$

- **Appearance function** $\Phi : (f_{\Omega(x)}, n, d) \mapsto c$
  - Input: intermediate feature vector $f_{\Omega(x)}$, a point normal $n \in \mathbb{R}^3$, and a view direction $d \in \mathbb{R}^3$
  - Output: color $c \in \mathbb{R}^3_+$



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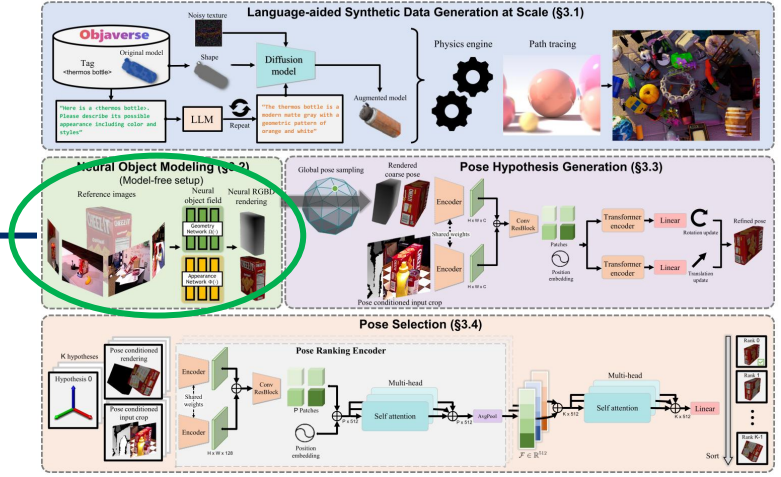

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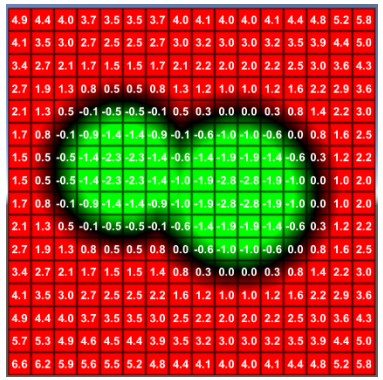
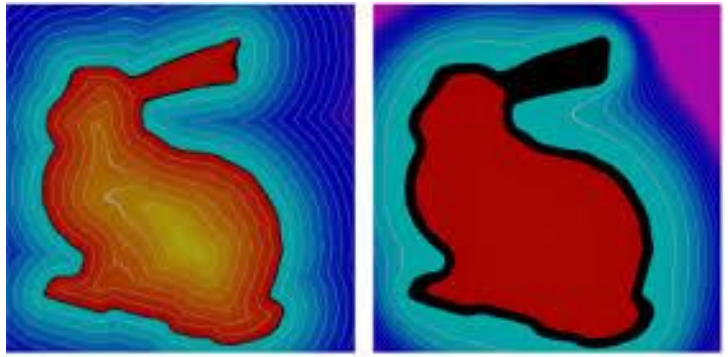
SDF의 gradient.
표면까지의 거리 = 최단 거리 (수직)

+

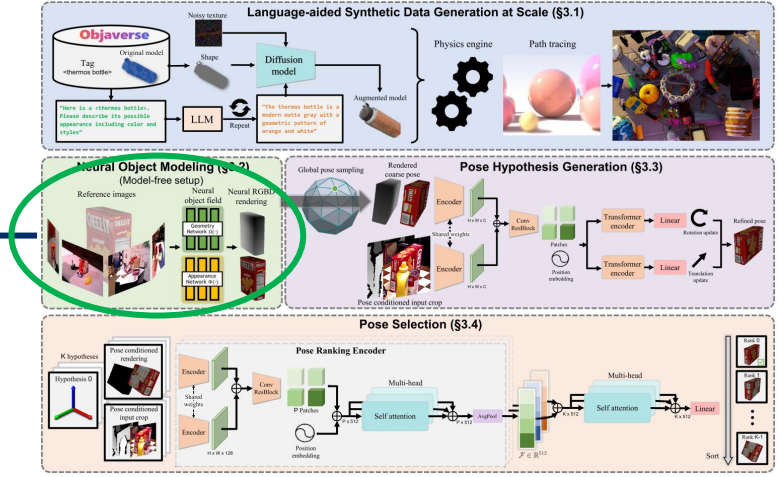
SDF : 점이 **물체 표면**으로부터 얼마나 떨어져 있는지 표현하는 방식
- **S > 0**, 점이 물체 표면 **밖**
- **S = 0**, 점이 물체 **표면**
- **S < 0**, 점이 물체 표면 **안**



4.9	4.4	4.0	3.7	3.5	3.5	3.7	4.0	4.1	4.0	4.0	4.1	4.4	4.8	5.2	5.8
4.1	3.5	3.0	2.7	2.5	2.5	2.7	3.0	3.2	3.0	3.0	3.2	3.5	3.9	4.4	5.0
3.4	2.7	2.1	1.7	1.5	1.5	1.7	2.1	2.2	2.0	2.0	2.2	2.5	3.0	3.6	4.3
2.7	1.9	1.3	0.8	0.5	0.5	0.8	1.3	1.2	1.0	1.0	1.2	1.6	2.2	2.9	3.6
2.1	1.3	0.5	-0.1	-0.5	-0.5	-0.1	0.5	0.3	0.0	0.0	0.3	0.8	1.4	2.2	3.0
1.7	0.8	-0.1	-0.9	-1.4	-1.4	-0.9	-0.1	-0.6	-1.0	-1.0	-0.6	0.0	0.8	1.6	2.5
1.5	0.5	-0.5	-1.4	-2.3	-2.3	-1.4	-0.6	-1.4	-1.9	-1.9	-1.4	-0.6	0.3	1.2	2.2
1.5	0.5	-0.5	-1.4	-2.3	-2.3	-1.4	-1.0	-1.9	-2.8	-2.8	-1.9	-1.0	0.0	1.0	2.0
1.7	0.8	-0.1	-0.9	-1.4	-1.4	-0.9	-1.0	-1.9	-2.8	-2.8	-1.9	-1.0	0.0	1.0	2.0
2.1	1.3	0.5	-0.1	-0.5	-0.5	-0.1	-0.6	-1.4	-1.9	-1.9	-1.4	-0.6	0.3	1.2	2.2
2.7	1.9	1.3	0.8	0.5	0.5	0.8	0.0	-0.6	-1.0	-1.0	-0.6	0.0	0.8	1.6	2.5
3.4	2.7	2.1	1.7	1.5	1.5	1.4	0.8	0.3	0.0	0.0	0.3	0.8	1.4	2.2	3.0
4.1	3.5	3.0	2.7	2.5	2.5	2.2	1.6	1.2	1.0	1.0	1.2	1.6	2.2	2.9	3.6
4.9	4.4	4.0	3.7	3.5	3.5	3.0	2.5	2.2	2.0	2.0	2.2	2.5	3.0	3.6	4.3
5.7	5.3	4.9	4.6	4.5	4.4	3.9	3.5	3.2	3.0	3.0	3.2	3.5	3.9	4.4	5.0
6.6	6.2	5.9	5.6	5.5	5.2	4.8	4.4	4.1	4.0	4.0	4.1	4.4	4.8	5.2	5.8



Method



3.2. Neural Object Modeling (For model-free)

Field Learning. **(Appearance function, Geometry function)**

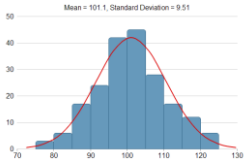
- **Appearance function**
  - volumetric rendering over **truncated near-surface regions**

z(r): depth value of ray from depth image
α(Hyperparameter): softness of the distribution.

$$c(r) = \int_{z(r)-\lambda}^{z(r)+0.5\lambda} w(x_i)\Phi(f_{\Omega(x_i)}, n(x_i), d(x_i))\, dt, \quad (1)$$

$$w(x_i) = \frac{1}{1+e^{-\alpha\Omega(x_i)}} \frac{1}{1+e^{\alpha\Omega(x_i)}}, \quad (2)$$

bell-shaped probability density function that depends on SDF



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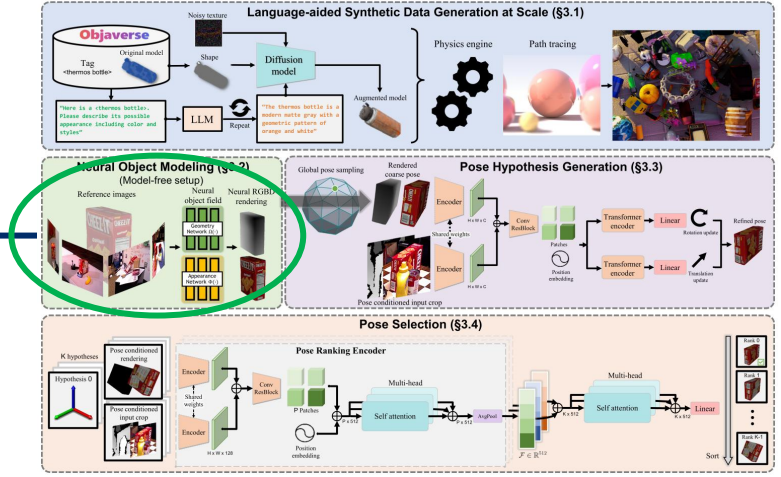

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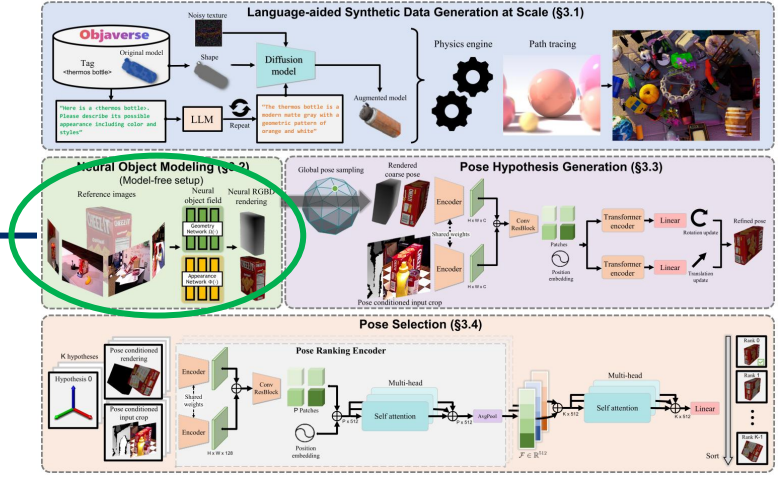
R: set of rays

$$\mathcal{L}_c = \frac{1}{|\mathcal{R}|} \sum_{r\in\mathcal{R}} \|c(r) - \bar{c}(r)\|_2 , \quad (3)$$

Method



3.2. Neural Object Modeling (For model-free)

Field Learning. **(Appearance function, Geometry function)**

- **Geometry function**
  - hybrid SDF model by dividing into **empty space** and **near-surface**.

$$\mathcal{L}_e = \frac{1}{|\mathcal{X}_e|} \sum_{x \in \mathcal{X}_e} |\Omega(x) - \lambda|, \qquad (4)$$

➔ 빈공간의 점과 표면과의 거리는 최소 λ로 지정.
(표면과의 거리의 GT가 없으니, 최소한 λ이상 떨어져있다는 정보)

$$\mathcal{L}_s = \frac{1}{|\mathcal{X}_s|} \sum_{x \in \mathcal{X}_s} (\underline{\Omega(x) + d_x} - d_D)^2, \qquad (5)$$

점과 표면과의 거리 + ray origin과 점의 거리= Depth

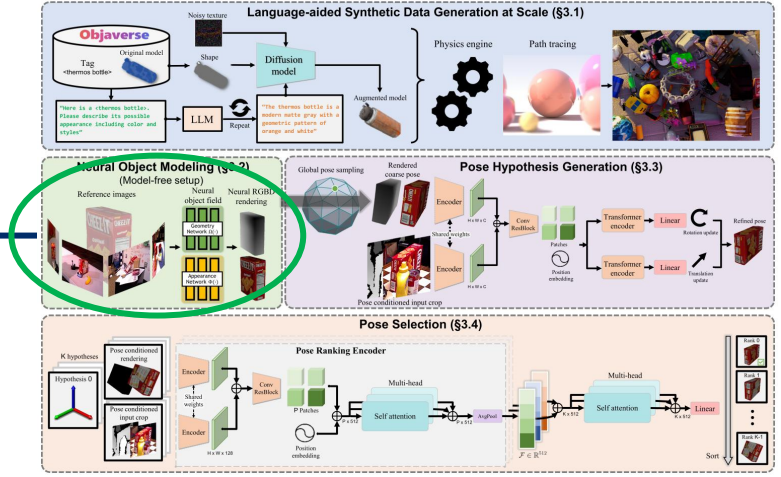
➔ SDF가 정확한 Depth를 예측하도록 Loss

$$\mathcal{L}_{eik} = \frac{1}{|\mathcal{X}_s|} \sum_{x \in \mathcal{X}_s} (\|\nabla \Omega(x)\|_2 - 1)^2, \qquad (6)$$

➔ 1만큼 움직이면 거리가 1만큼 변하도록
즉, 수직으로 이동 (gradient = 1)

X_e: sampled Point set in **empty space**
X_s: sampled point set in **near-surface**
X: sampled 3d point
d_x: distance from **ray origin** to **point**
d_D: observed depth point

Method



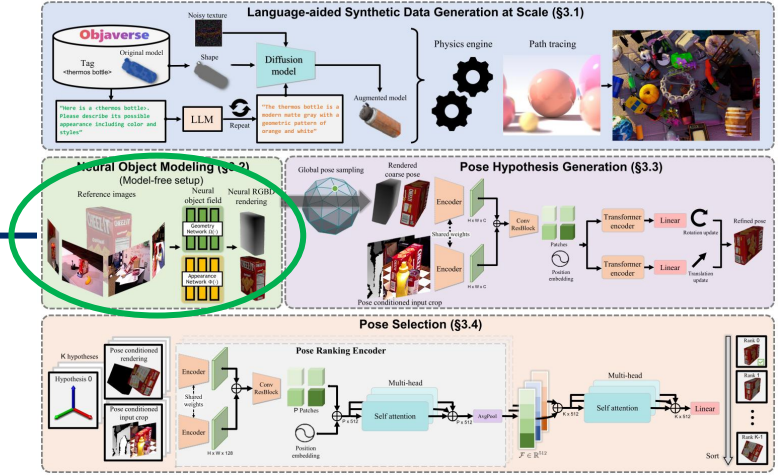
3.2. Neural Object Modeling (For model-free)

Field Learning. (Appearance function, Geometry function)

- Total Loss

$$\mathcal{L} = \underbrace{w_c\mathcal{L}_c}_{\text{Appearance function Loss}} + \underbrace{w_e\mathcal{L}_e + w_s\mathcal{L}_s + w_{eik}\mathcal{L}_{eik}}_{\text{Geometry function Loss}}. \tag{7}$$

Method



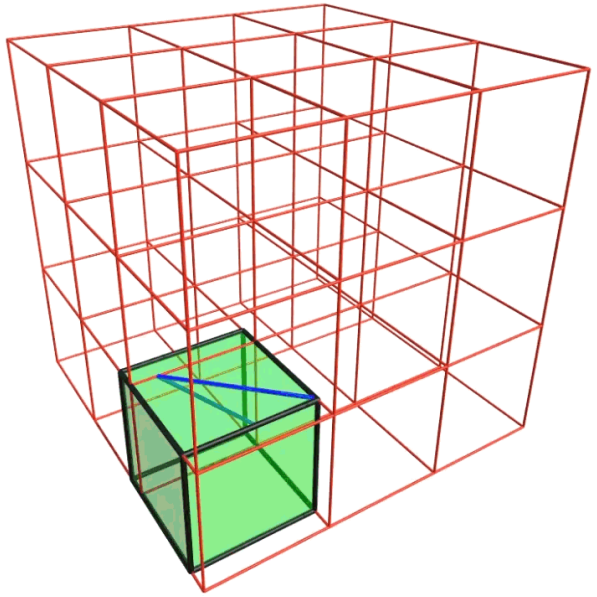
3.2. Neural Object Modeling (For model-free)

Rendering.

- Once trained, perform **marching cubes** to extract a **textured mesh**, combined **with color** projection.
- At inference, given an object pose, then **render the RGBD image** following the **rasterization** process.

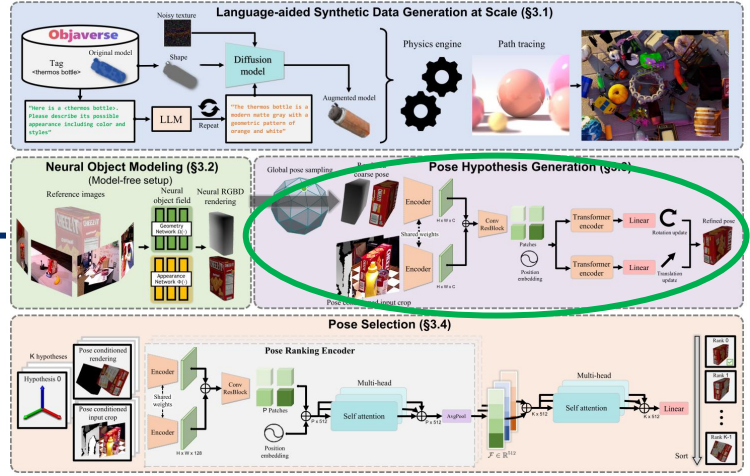
marching cubes :

Neural field → 3D Mesh로 만드는 Algorithm.
Model-based의 3D CAD모델과 같은 역할



1. 3D 공간을 작은 정육면체(cube) 격자로 나눈다.
2. 순회하면서 8개 꼭지점에서 SDF 값을 구한다.
3. SDF의 부호가 바뀌는(양수↔음수) 변(edge)의 교차점들을 찾는다.
4. 점들을 삼각형으로 연결한다.

Method



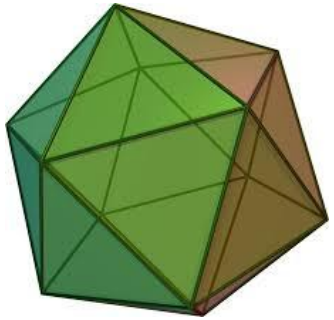
3.3. Pose Hypothesis Generation

(coarse) Pose Initialization.

- Given the **RGBD** image, **object** is **detected** using **Mask RCNN** or **CNOS**.

Translation Initialization

- using the **3D point** located at the **median depth** within **2D bounding box**.



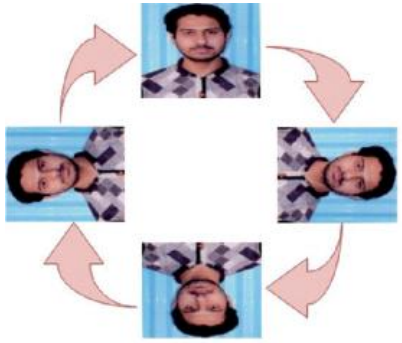
꼭지점 12개
변의 중점 30개
(level=1)

Rotation initialization

- uniformly sample Ns(42) **viewpoints** from an **icosphere**.
- These **camera poses** are further **augmented** with Ni(12) discretized **in-plane rotations**.

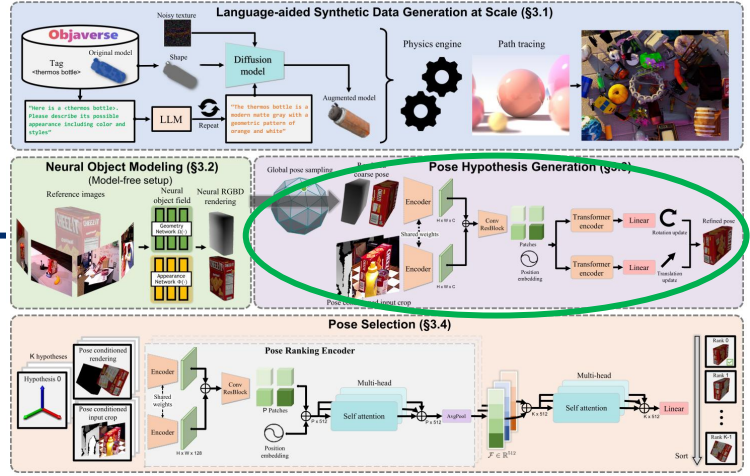
➔ resulting in **Ns(42) · Ni(12) = 504** pose initializations

360° / 12 = 30° 간격
[30, 60, 90, …360]



(a) In-plane rotation

Method



3.3. Pose Hypothesis Generation

Pose Refinement.

- coarse pose initializations need **refinement**.

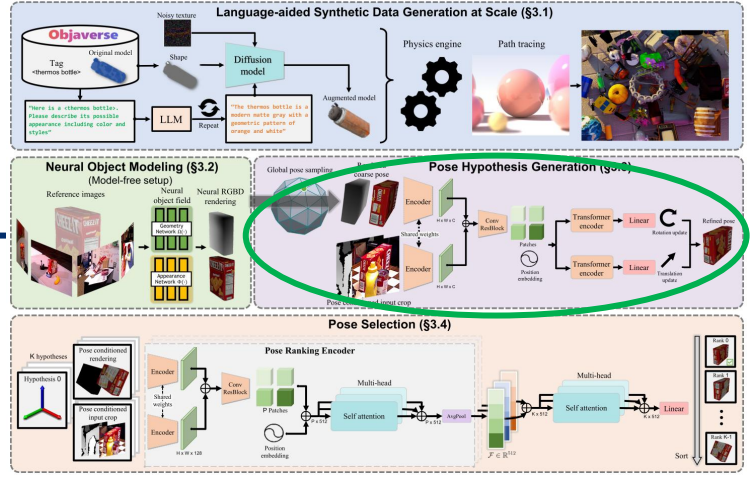
Pose Refinement Network

- Input: **rendered image** on coarse **pose**, and a **crop of the RGB-D** from camera
- Output: updated pose

Crop of the RGB-D

1. 물체 중심의 **3D 좌표**를 intrinsics로 **2D**에 **projection**해서 **crop 중심**을 구한다.
2. 물체의 **3D diameter**를 **2D**에 **projection**해서 **crop 크기**를 구한다

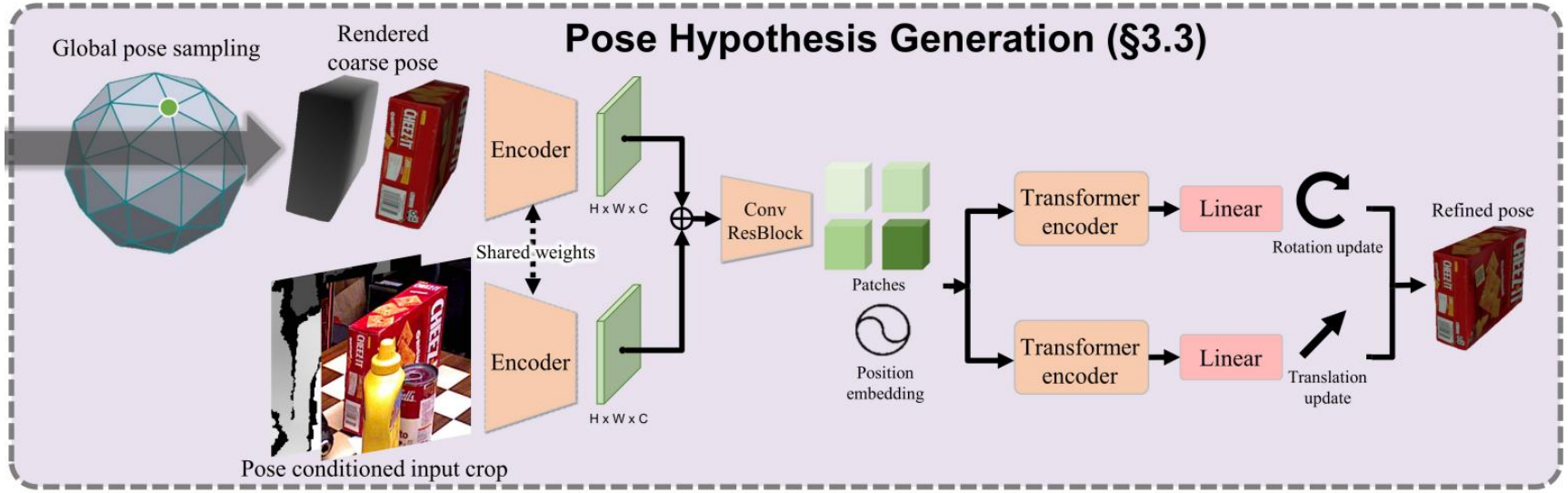
Method



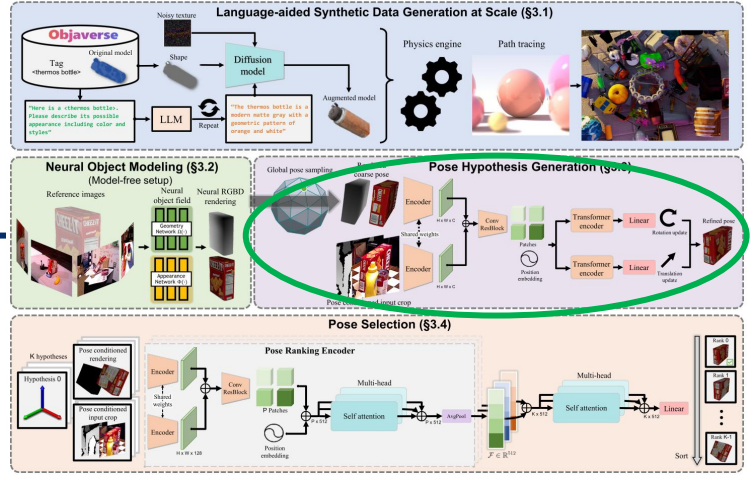
3.3. Pose Hypothesis Generation

Pose Refinement Network

1. **Extract feature** maps from the two RGBD input with a **shared CNN encoder**.
2. The feature maps are **concatenated**, **fed** into **CNN** blocks with **residual connection.**
3. **Tokenized** by dividing into **patches** with **position embedding**.
4. Finally, the network **predicts** $\Delta \boldsymbol{t} \in \mathbb{R}^3$ and $\Delta \boldsymbol{R} \in \mathbb{SO}(3)$.
5. Repeat.



Method



3.3. Pose Hypothesis Generation

Disentangled Representation (not homogeneous)

- To removes the dependency between orientation update and translation update.

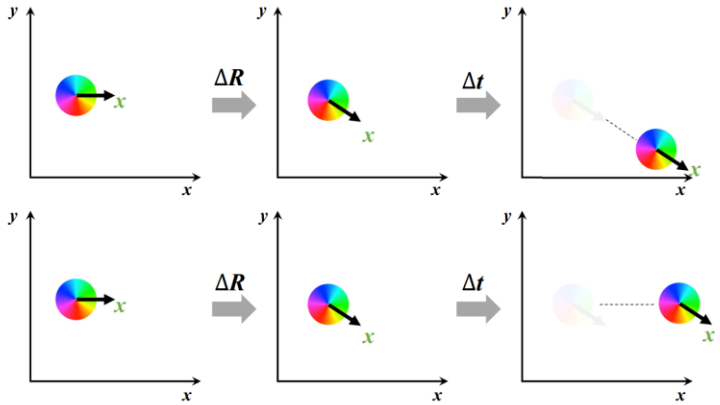


Figure 10. Illustration of disentangled representation for pose updates.

Homogeneous → rotation affects the translation

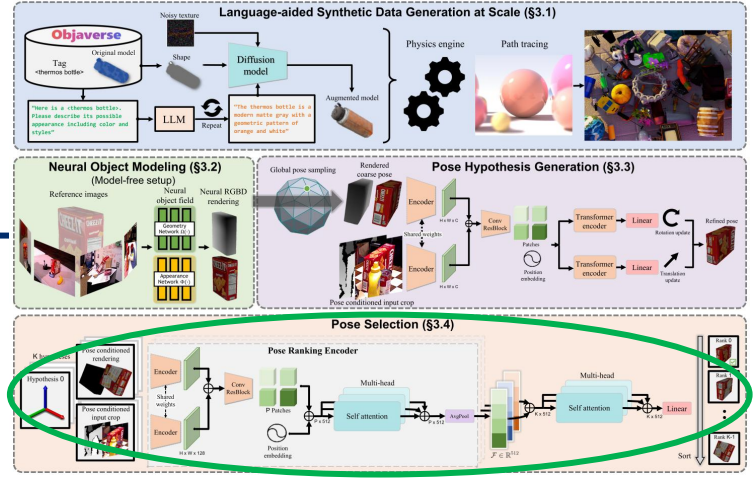
Disentangled → resolves the dependency → stabilizes training

$$\boldsymbol{t}^{+} = \boldsymbol{t} + \Delta\boldsymbol{t} \tag{8}$$

$$\boldsymbol{R}^{+} = \Delta\boldsymbol{R} \otimes \boldsymbol{R}, \tag{9}$$

$$\mathcal{L}_{\text{refine}} = w_1 \left\|\Delta\boldsymbol{t} - \Delta\bar{\boldsymbol{t}}\right\|_2 + w_2 \left\|\Delta\boldsymbol{R} - \Delta\bar{\boldsymbol{R}}\right\|_2, \tag{10}$$

Method



3.4. Pose Selection

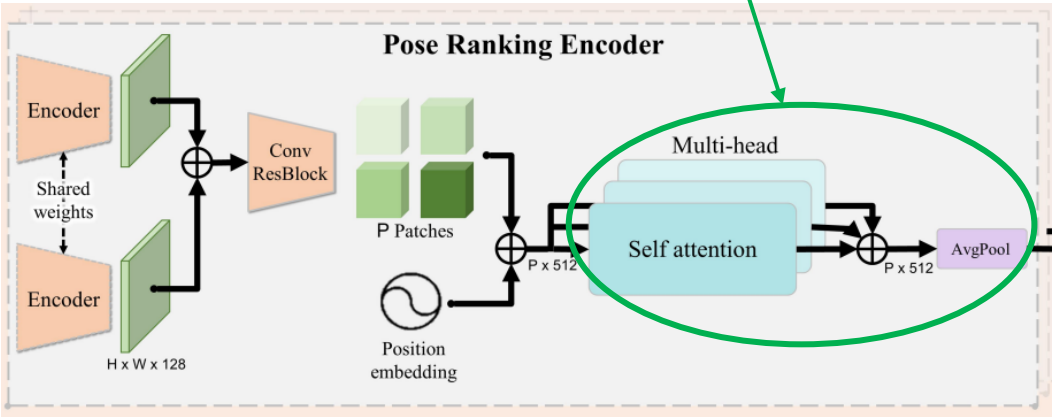
Hierarchical(two-level) Comparison.

First level,
for each **pose hypothesis**, the **rendered image** is **compared** against the **cropped input observation**.

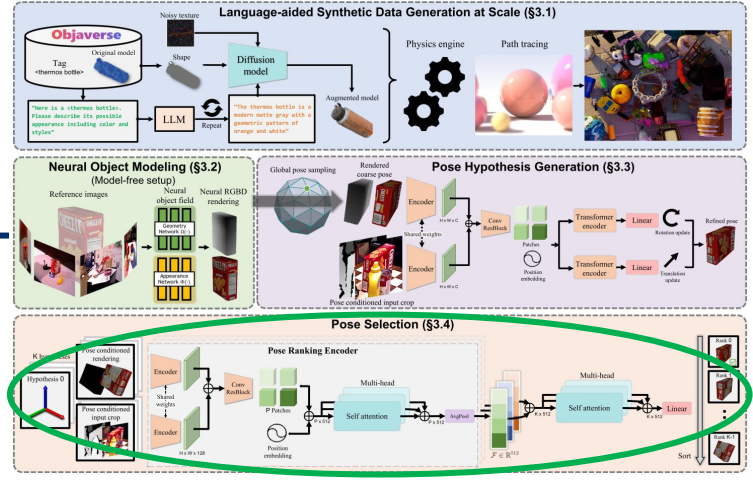
- **Extract feature** maps from the two RGBD input with a **shared CNN encoder**.
- The feature maps are **concatenated**, **fed** into **CNN** blocks with **residual connection.**
- **Tokenized** by dividing into **patches** with **position embedding**.

same architecture as in Sec. 3.3

- **forwarded** to the **multi-head self-attention** module.
- **average pooling** to output a **feature embedding** $\mathcal{F} \in \mathbb{R}^{512}$



Method

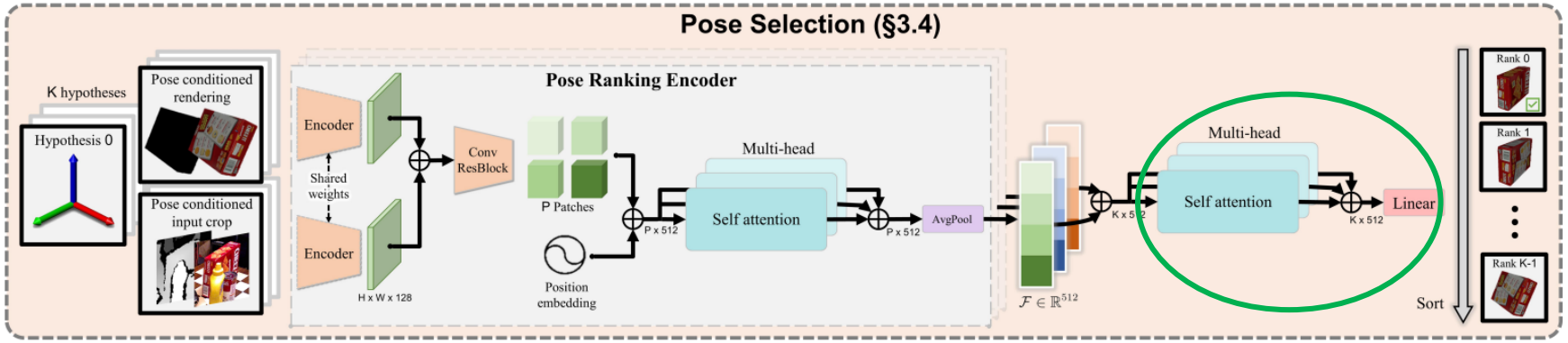


3.4. Pose Selection

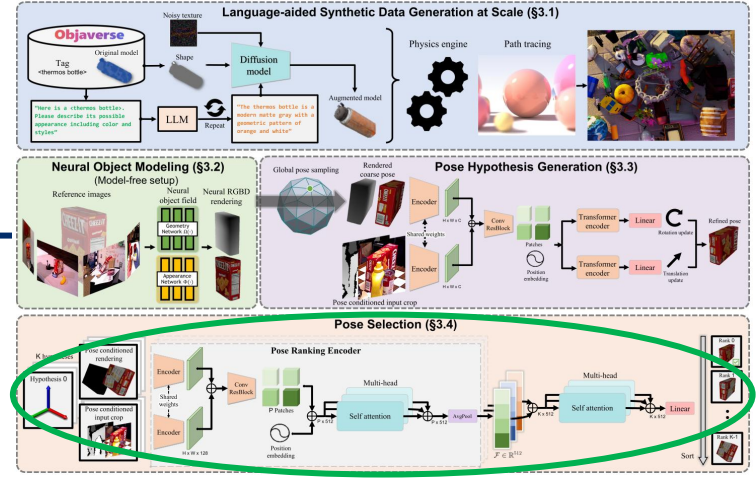
Hierarchical(two-level) Comparison.

Second level,

- Multi-head self-attention is performed on the concatenated feature embedding $\mathbf{F} = [\mathcal{F}_0, \ldots, \mathcal{F}_{K-1}]^\top \in \mathbb{R}^{K \times 512}$
- Then, linearly projected to the scores $\mathbf{S} \in \mathbb{R}^K$



Method



3.4. Pose Selection

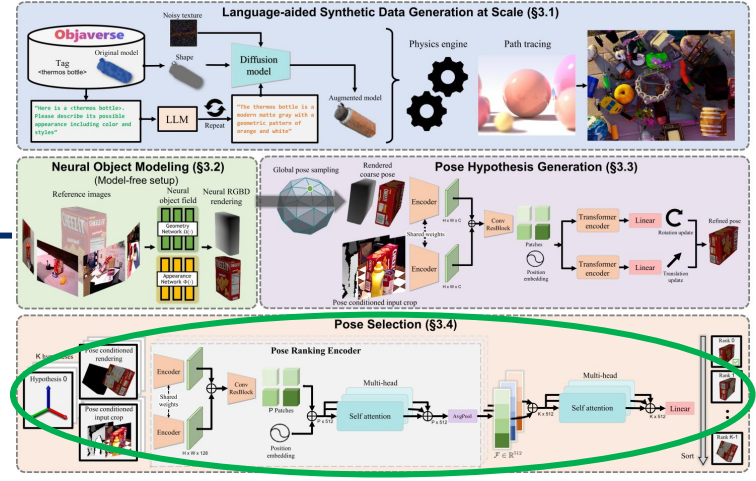
Contrast Validation.
propose a pose-conditioned **triplet loss**

$$\mathcal{L}(i^+, i^-) = \max(\mathbf{S}(i^-) - \mathbf{S}(i^+) + \alpha, 0), \qquad (11)$$

➔ 좋은 점수가 나쁜 점수보다 α 이상 높아야 Loss = 0

α: contrastive margin
i − , i + : negative and positive pose samples, which are determined by computing the ADD metric using GT
S(i-), S(i+) : score of samples

Method



3.4. Pose Selection

Contrast Validation.

i-, i+ 모두 정답과 거리가 멀다면 **triplet loss**는 의미가 없어진다.
GT와 가까운 positive sample만 사용

$$\mathbb{V}^{+} = \{i \,:\, D(\boldsymbol{R}_i, \bar{\boldsymbol{R}}) < d\} \tag{12}$$

$$\mathbb{V}^{-} = \{0, 1, 2, \ldots, K-1\} \tag{13}$$

$$\mathcal{L}_{\text{rank}} = \sum_{i^+, i^-} \mathcal{L}(i^+, i^-) \tag{14}$$

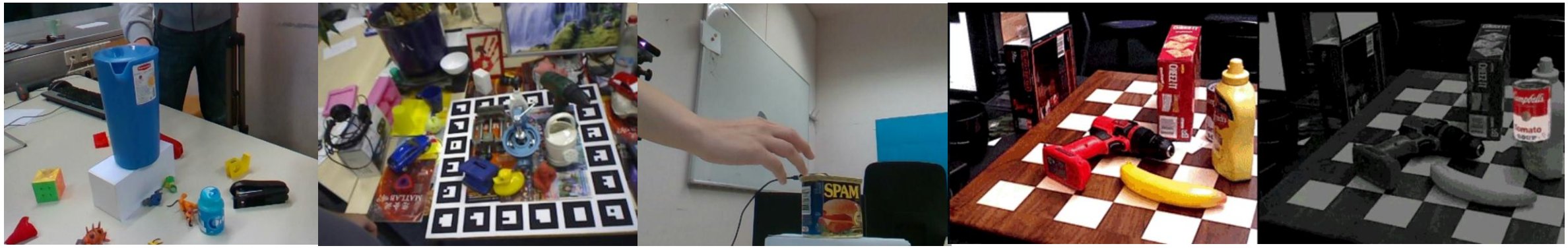
R: rotation of hypothesis and GT
D(): geodesic distance between rotations

Experiment

Datasets

다양한 시나리오, 다양한 물체 속성을 가진 Datasets

YCBInEOAT
LINEMOD
Occluded-LINEMOD
YCB-Video
T-LESS



Experiment

Quantitative Comparison
(Model-free)

각 분야의 SOTA모델과 비교

- PREDATOR: 3D Point Cloud Matching
- LoFTR: 2D Feature Matching
- FS6D-DPM: Matching 기반 RGB-D Model-free 6D OPE

	PREDATOR [28]		LoFTR [57]		FS6D-DPM [22]		Ours	
Ref. images	16		16		16		16	
Finetune-free	✓		✓		✗		✓	
Metrics	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD
002_master_chef_can	73.0	17.4	87.2	50.6	92.6	36.8	96.9	91.3
003_cracker_box	41.7	8.3	71.8	25.5	83.9	24.5	97.5	96.2
004_sugar_box	53.7	15.3	63.9	13.4	95.1	43.9	97.5	87.2
005_tomato_soup_can	81.2	44.4	77.1	52.9	93.0	54.2	97.6	93.3
006_mustard_bottle	35.5	5.0	84.5	59.0	97.0	71.1	98.4	97.3
007_tuna_fish_can	78.2	34.2	72.6	55.7	94.5	53.9	97.7	73.7
008_pudding_box	73.5	24.2	86.5	68.1	94.9	79.6	98.5	97.0
009_gelatin_box	81.4	37.5	71.6	45.2	98.3	32.1	98.5	97.3
010_potted_meat_can	62.0	20.9	67.4	45.1	87.6	54.9	96.6	82.3
011_banana	57.7	9.9	24.2	1.6	94.0	69.1	98.1	95.4
019_pitcher_base	83.7	18.1	58.7	22.3	91.1	40.4	97.9	96.6
021_bleach_cleanser	88.3	48.1	36.9	16.7	89.4	44.1	97.4	93.3
024_bowl	73.2	17.4	32.7	1.4	74.7	0.9	94.9	89.7
025_mug	84.8	29.5	47.3	23.6	86.5	39.2	96.2	75.8
035_power_drill	60.6	12.3	18.8	1.3	73.0	19.8	98.0	96.3
036_wood_block	70.5	10.0	49.9	1.4	94.7	27.9	97.4	94.7
037_scissors	75.5	25.0	32.3	14.6	74.2	27.7	97.8	95.5
040_large_marker	81.8	38.9	20.7	8.4	97.4	74.2	98.6	96.5
051_large_clamp	83.0	34.4	24.1	11.2	82.7	34.7	96.9	92.7
052_extra_large_clamp	72.9	24.1	15.0	1.8	65.7	10.1	97.6	94.1
061_foam_brick	79.2	35.5	59.4	31.4	95.7	45.8	98.1	93.4
MEAN	71.0	24.3	52.5	26.2	88.4	42.1	**97.4**	**91.5**

Table 1. Model-free pose estimation results measured by AUC of ADD and ADD-S on YCB-Video dataset. "Finetuned" means the method was fine-tuned with group split of object instances on the testing dataset, as introduced by [22].

Test Data로 추가 학습
이유:
일반화 성능이 떨어져서 성능 ↓

Experiment

Quantitative Comparison
(Model-free)

각 분야의 SOTA모델과 비교

- PREDATOR: 2D Feature Matching
- LoFTR: 3D Point Cloud Matching
- FS6D-DPM: Matching 기반 RGB-D Model-free 6D OPE

ADD (Average 3D Distance):
예측한 점과 GT점과 1:1 매칭 거리.
회전/이동(6D Pose) 맞아야 높은 점수를 받는다.

ADD-S (Symmetric ADD):
Symmetric 물체를 고려한 더 관대한 Metric.
GT와 상관없이, 예측한 점과 물리적으로 가장 가까운 점과의 거리.

	PREDATOR [28]		LoFTR [57]		FS6D-DPM [22]		Ours	
Ref. images	16		16		16		16	
Finetune-free	✓		✓		✗		✓	
Metrics	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD
002_master_chef_can	73.0	17.4	87.2	50.6	92.6	36.8	96.9	91.3
003_cracker_box	41.7	8.3	71.8	25.5	83.9	24.5	97.5	96.2
004_sugar_box	53.7	15.3	63.9	13.4	95.1	43.9	97.5	87.2
005_tomato_soup_can	81.2	44.4	77.1	52.9	93.0	54.2	97.6	93.3
006_mustard_bottle	35.5	5.0	84.5	59.0	97.0	71.1	98.4	97.3
007_tuna_fish_can	78.2	34.2	72.6	55.7	94.5	53.9	97.7	73.7
008_pudding_box	73.5	24.2	86.5	68.1	94.9	79.6	98.5	97.0
009_gelatin_box	81.4	37.5	71.6	45.2	98.3	32.1	98.5	97.3
010_potted_meat_can	62.0	20.9	67.4	45.1	87.6	54.9	96.6	82.3
011_banana	57.7	9.9	24.2	1.6	94.0	69.1	98.1	95.4
019_pitcher_base	83.7	18.1	58.7	22.3	91.1	40.4	97.9	96.6
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024_bowl	73.2	17.4	32.7	1.4	74.7	0.9	94.9	89.7
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035_power_drill	60.6	12.3	18.8	1.3	73.0	19.8	98.0	96.3
036_wood_block	70.5	10.0	49.9	1.4	94.7	27.9	97.4	94.7
037_scissors	75.5	25.0	32.3	14.6	74.2	27.7	97.8	95.5
040_large_marker	81.8	38.9	20.7	8.4	97.4	74.2	98.6	96.5
051_large_clamp	83.0	34.4	24.1	11.2	82.7	34.7	96.9	92.7
052_extra_large_clamp	72.9	24.1	15.0	1.8	65.7	10.1	97.6	94.1
061_foam_brick	79.2	35.5	59.4	31.4	95.7	45.8	98.1	93.4
MEAN	71.0	24.3	52.5	26.2	88.4	42.1	**97.4**	**91.5**

Table 1. Model-free pose estimation results measured by AUC of ADD and ADD-S on YCB-Video dataset. "Finetuned" means the method was fine-tuned with group split of object instances on the testing dataset, as introduced by [22].

PREDATOR, LoFTR은
매칭 결과 + PnP/SVD 로
6D Pose 계산하여 비교

Outperform.
➔ FoundationPose의 Render-and-compare가 기존 Matching 방법을 앞선다.

Experiment

Quantitative Comparison
(Model-free)

Method	Modality	Finetune-free	Ref. images	Objects														Avg.
				ape	benchwise	cam	can	cat	driller	duck	eggbox	glue	holepuncher	iron	lamp	phone		
Gen6D [40]	RGB	✗	200	-	77	66.1	-	60.7	67.4	40.5	95.7	87.2	-	-	-	-		-
Gen6D* [40]	RGB	✓	200	-	62.1	45.6	-	40.9	48.8	16.2	-	-	-	-	-	-		-
OnePose [58]	RGB	✓	200	11.8	92.6	88.1	77.2	47.9	74.5	34.2	71.3	37.5	54.9	89.2	87.6	60.6		63.6
OnePose++ [19]	RGB	✓	200	31.2	97.3	88.0	89.8	70.4	92.5	42.3	99.7	48.0	69.7	97.4	97.8	76.0		76.9
LatentFusion [51]	RGBD	✓	16	88.0	92.4	74.4	88.8	94.5	91.7	68.1	96.3	94.9	82.1	74.6	94.7	91.5		87.1
FS6D [22]	RGBD	✗	16	74.0	86.0	88.5	86.0	98.5	81.0	68.5	**100.0**	99.5	97.0	92.5	85.0	99.0		88.9
FS6D [22] + ICP	RGBD	✗	16	78.0	88.5	91.0	89.5	97.5	92.0	75.5	99.5	99.5	96.0	87.5	97.0	97.5		91.5
Ours	RGBD	✓	16	**99.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**99.4**	**100.0**	**100.0**	**99.9**	**100.0**	**100.0**	**100.0**		**99.9**

Table 2. Model-free pose estimation results measured by ADD-0.1d on LINEMOD dataset. Gen6D* [40] represents the variation without fine-tuning.

성공률,
오차가 10%보다 적은 비율

Experiment

Quantitative Comparison
(Model-based)

Method	Unseen objects	Dataset LM-O	T-LESS	YCB-V	Mean
SurfEmb [15] + ICP	✗	75.8	82.8	80.6	79.7
OSOP [55] + ICP	✓	48.2	-	57.2	-
(PPF, Sift) + Zephyr [48]	✓	59.8	-	51.6	-
MegaPose-RGBD [32]	✓	58.3	54.3	63.3	58.6
OVE6D [2]	✓	49.6	52.3	-	-
GCPose [76]	✓	65.2	67.9	-	-
Ours	✓	**78.8**	**83.0**	**88.0**	**83.3**

Table 3. Model-based pose estimation results measured by AR score on representative BOP datasets. All methods use the RGBD modality.

Average Recall,
로봇 조작(MSSD), 증강현실(MSPD), 심한 가림(VSD) - 3개 평가 지표의 **평균**

Experiment

Qualitative Comparison
(Model-free)

Figure 5. Qualitative comparison of pose estimation on LINEMOD dataset under the model-free setup. Images are cropped and zoomed-in for better visualization.

Experiment

Quantitative Comparison
(Pose Tracking)

		se(3)-TrackNet [67]	RGF [29]	Bundle-Track [66]	Bundle-SDF [71]	Wüthrich [72]	Ours	Ours†
Properties	Novel object	✗	✓	✓	✓	✓	✓	✓
	Initial pose	GT	GT	GT	GT	GT	GT	Est.
cracker_box	ADD-S	94.06	55.44	89.41	90.63	88.13	**95.10**	**94.92**
	ADD	90.76	34.78	85.07	85.37	79.00	**91.32**	**91.54**
bleach_cleanser	ADD-S	94.44	45.03	94.72	94.28	68.96	**95.96**	**96.36**
	ADD	89.58	29.40	89.34	87.46	61.47	**91.45**	**92.63**
sugar_box	ADD-S	94.80	16.87	90.22	93.81	92.75	**96.67**	**96.61**
	ADD	92.43	15.82	85.56	88.62	86.78	**94.14**	**93.96**
tomato_soup_can	ADD-S	96.95	26.44	95.13	95.24	93.17	**96.58**	**96.54**
	ADD	93.40	15.13	86.00	83.10	63.71	**91.71**	**91.85**
mustard_bottle	ADD-S	97.92	60.17	95.35	95.75	95.31	**97.89**	**97.77**
	ADD	97.00	56.49	92.26	89.87	91.31	**96.34**	**95.95**
All	ADD-S	95.53	39.90	92.53	93.77	89.18	**96.42**	**96.40**
	ADD	92.66	29.98	87.34	86.95	78.28	**93.09**	**93.22**

Table 4. Pose tracking results of RGBD methods measured by AUC of ADD and ADD-S on YCBInEOAT dataset. Ours† represents our unified pipeline that uses the pose estimation module for pose initialization.

Approach	DeepIM [36]		se(3)-TrackNet [67]		PoseRBPF [8] + SDF		Wüthrich [72]		RGF [29]		ICG [56]		Ours		Ours†	
Initial pose	GT		GT		PoseCNN		GT		GT		GT		GT		GT	
Re-initialization	Yes (290)		No		Yes (2)		No		No		No		No		No	
Novel object	✗		✗		✗		✓		✓		✓		✓		✓	
Object setup	Model-based		Model-based		Model-based		Model-based		Model-based		Model-based		Model-based		Model-free	
Metric	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S	ADD	ADD-S
002_master_chef_can	89.0	93.8	93.9	96.3	89.3	96.7	55.6	90.7	46.2	90.2	66.4	89.7	93.6	97.0	91.2	96.9
003_cracker_box	88.5	93.0	96.5	97.2	96.0	97.1	96.4	97.2	57.0	72.3	82.4	92.1	96.9	97.8	96.2	97.5
004_sugar_box	94.3	96.3	97.6	98.1	94.0	96.4	97.1	97.9	50.4	72.7	96.1	98.4	96.9	98.2	94.5	97.4
005_tomato_soup_can	89.1	93.2	95.0	97.2	87.2	95.2	64.7	89.5	72.4	91.6	73.2	97.3	96.3	98.1	94.3	97.9
006_mustard_bottle	92.0	95.1	95.8	97.4	98.3	98.5	97.1	98.0	87.7	98.2	96.2	98.4	97.3	98.4	97.3	98.5
007_tuna_fish_can	92.0	96.4	86.5	91.1	86.8	93.6	69.1	93.3	28.7	52.9	73.2	95.8	96.9	98.5	84.0	97.8
008_pudding_box	80.1	88.3	97.9	98.4	60.9	87.1	96.8	97.9	12.7	18.0	73.8	88.9	97.8	98.5	96.9	98.5
009_gelatin_box	92.0	94.4	97.8	98.4	98.2	98.6	97.5	98.4	49.1	70.7	97.2	98.8	97.7	98.5	97.6	98.5
010_potted_meat_can	78.0	88.9	77.8	84.2	76.4	83.5	83.7	86.7	44.1	45.6	93.3	97.3	95.1	97.7	94.8	97.5
011_banana	81.0	90.5	94.9	97.2	92.8	97.7	86.3	96.1	93.3	97.7	95.6	98.4	96.4	98.4	95.6	98.1
019_pitcher_base	90.4	94.7	96.8	97.5	97.7	98.1	97.3	97.7	97.9	98.2	97.0	98.8	96.7	98.0	96.8	98.0
021_bleach_cleanser	81.7	90.5	95.9	97.2	95.9	97.0	95.2	97.2	95.9	97.3	92.6	97.5	95.5	97.8	94.7	97.5
024_bowl	38.8	90.6	80.9	94.5	34.0	93.0	30.4	97.2	24.2	82.4	74.4	98.4	95.2	97.6	90.5	95.3
025_mug	83.2	92.0	91.5	96.9	86.9	96.7	83.2	93.3	60.0	71.2	95.6	98.5	95.6	97.9	91.5	96.1
035_power_drill	85.4	92.3	96.4	97.4	97.8	98.2	97.1	97.8	97.9	98.3	96.7	98.5	96.9	98.2	96.3	97.9
036_wood_block	44.3	75.4	95.2	96.7	37.8	93.6	95.5	96.9	45.7	62.5	93.5	97.2	93.2	97.0	92.9	97.0
037_scissors	70.3	84.5	95.7	97s	72.7	85.5	4.2	16.2	20.9	38.6	93.5	97.3	94.8	97.5	95.5	97.8
040_large_marker	80.4	91.2	92.2	96.0	89.2	97.3	35.6	53.0	12.2	18.9	88.5	97.8	96.9	98.6	96.6	98.6
051_large_clamp	73.9	84.1	94.7	96.9	90.1	95.5	61.2	72.3	62.8	80.1	91.8	96.9	93.6	97.3	92.5	96.7
052_extra_large_clamp	49.3	90.3	91.7	95.8	84.4	94.1	93.7	96.6	67.5	69.7	85.9	94.3	94.4	97.5	93.4	97.3
061_foam_brick	91.6	95.5	93.7	96.7	96.1	98.3	96.8	98.1	70.0	86.5	96.2	98.5	97.9	98.6	96.8	98.3
All Frames	82.3	91.9	93.0	95.7	87.5	95.2	78.0	90.2	59.2	74.3	86.4	96.5	**96.0**	**97.9**	**93.7**	**97.5**

Table 5. Pose tracking results of RGBD methods measured by AUC of ADD and ADD-S on YCB-Video dataset. Ours† represents our method under the model-free setup with reference images.

Experiment

Ablation Study

Model-free, 16 ref images

Transformer → convolution and linear layers

Contrast loss → InfoNCE

	ADD	ADD-S
Ours (proposed)	91.52	97.40
W/o LLM texture augmentation	90.83	97.38
W/o transformer	90.77	97.33
W/o hierarchical comparison	89.05	96.67
Ours-InfoNCE	89.39	97.29

Table 6. Ablation study of critical design choices.

Experiment

Ablation Study

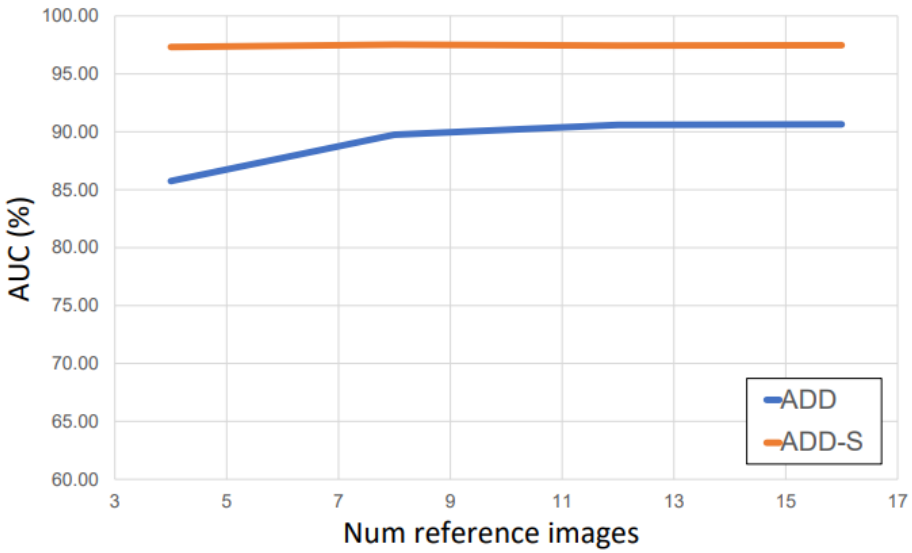

Figure 6. Effects of number of reference images.

Effects of number of reference images

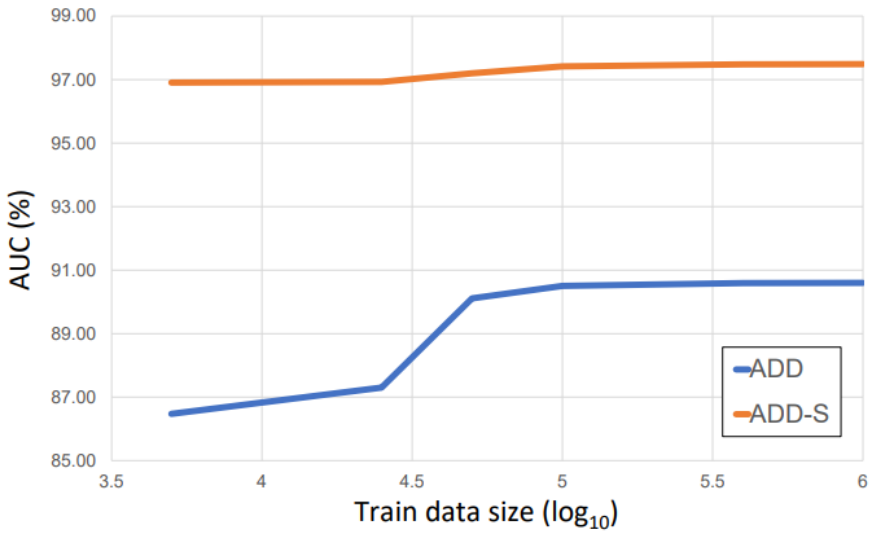

Figure 7. Effects of training data size.

Training data scaling law

Experiment

Failure Mode

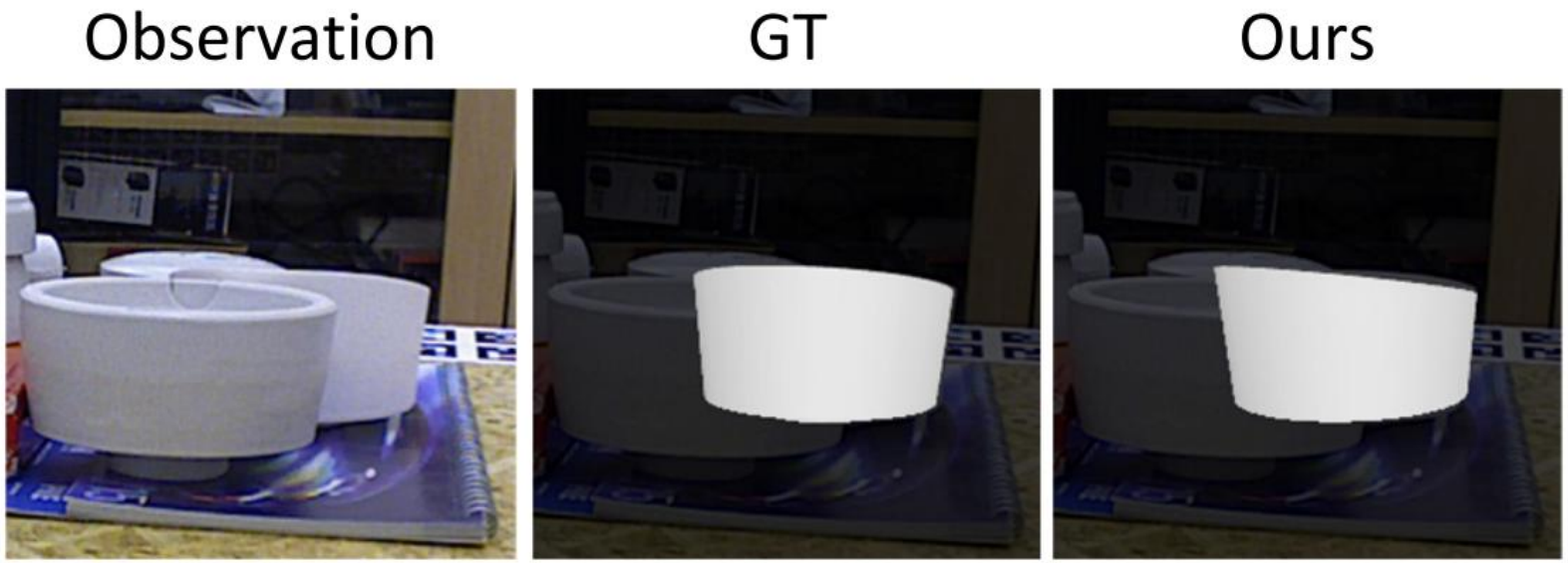

Figure 11. Failure mode. Under the combination of multiple challenges including texture-less, severe occlusion, and limited edge cues, our method fails to estimate the correct orientation.

Conclusion

Contribution

- present a **unified framework** for both **pose estimation** and **tracking** for **novel objects**, supporting both **model-based** and **model-free** setups.
- propose a **LLM-aided synthetic data generation pipeline**.
- **novel** design of **transformer-based network architectures** and **contrastive learning formulation** (→ strong generalization when trained solely on synthetic data)
- **outperforms** existing methods and achieves **comparable** results to **instance-level methods**.

Conclusion

Limitation

- **Relies** on **external 2D detection Model**.
- **Vulnerability** to **Texture-less, Symmetric**, severe **occlusion**, and **limited edge** cues.
- There is **no mechanism** to **auto-recover** when tracking fails.
- The '**Model-free**' approach **also needs** several reference **images**.